Randomized Trees with Conditional Inference

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This talk is based on

Classification Trees with Valid Inference via the Exponential Mechanism

`arxiv.org/abs/2511.15068`

joint work with my advisor

Snigdha Panigrahi

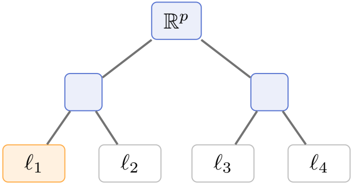
Decision trees: interpretable, non-linear, and "algorithmic"

A tree recursively partitions the predictor space into interpretable, rule-based regions.

Fit a simple model on each leaf $\ell_{m'}$:

$$\textbf{regression:}\ \ Y_i \mid X_i \in \ell_{m'} \ \sim\ N(\mu_{m'}, \sigma^2)$$

$$\textbf{classification:}\ \ Y_i \mid X_i \in \ell_{m'} \ \sim\ \text{Bernoulli}(\pi_{m'})$$



$\{\ell_1, \ldots, \ell_m\}$ partition $\mathbb{R}^p$

Breiman (2001): trees are *algorithmic* models.

Inference in an algorithmic model?

Classical workflow

define model ↓ observe data ↓ infer

Algorithmic workflow

examine data ↓ fit the tree ↓ infer?

same data: select the tree $\longrightarrow$ infer depends on the tree fitted!

Naive: coverage undercovers $\approx 60\%$ at nominal 90%

Data splitting: valid, but uses $n/2$ $\Rightarrow$ wider CIs, worse fits

Randomize the split selection: valid inference conditional on the observed tree fit, using full data.

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Sample the winner, don't declare it

Candidate splits $s' \in \mathcal{K}(\mathcal{P})$ at a parent region $\mathcal{P}$; gain $G_{\mathcal{P}}(s'; y)$.

Greedy

CART (Breiman et al., 1984); C4.5 (Quinlan, 2014)

$$s^* = \operatorname*{argmax}_{s' \in \mathcal{K}(\mathcal{P})} G_{\mathcal{P}}(s'; y)$$

hard argmax, discontinuous in y

Exponential mechanism

this work: **iCART**

$$\mathbb{P}(S = s \mid y) = \frac{\exp\left(\tau_{\mathcal{P}}^{-1} G_{\mathcal{P}}(s; y)\right)}{\sum_{s' \in \mathcal{K}(\mathcal{P})} \exp\left(\tau_{\mathcal{P}}^{-1} G_{\mathcal{P}}(s'; y)\right)}$$

one smooth softmax draw per node

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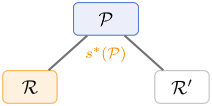

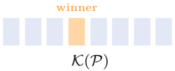

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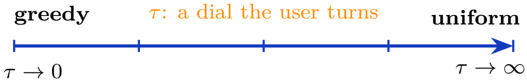

Zooming in on one leaf

$$\underbrace{Y = y}_{\text{data}} \xrightarrow{\text{fit: softmax splits}} \underbrace{\mathcal{T} = \mathbf{t}}_{\text{tree}} \xrightarrow{\text{zoom in}} \underbrace{L = \ell}_{\text{observed leaf}}$$

$$\boxed{p_1 = \mathcal{X}} \xrightarrow{S_1 = s_1} \boxed{p_2} \xrightarrow{S_2 = s_2} \cdots \longrightarrow \boxed{p_d} \xrightarrow{S_d = s_d} \boxed{\ell}$$

faded: rest of the tree

Selection event: the ancestral chain $\bigcap_{h \in [d]} \{S_h = s_h\}$ from root to ℓ.

T_ℓ leaf **statistic**: sample mean / sample proportion in ℓ

θ_ℓ leaf **parameter**, the target: subgroup mean μ_ℓ / rate π_ℓ

A_ℓ **rest of the data**: $y \equiv y(T_\ell, A_\ell)$; held fixed

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Conditioning on the fit: the correction factorizes

Condition on the ancestral chain that produced ℓ; the density of T_ℓ:

$$p\Big(t \,\Big|\, \textstyle\bigcap_h \{S_h = s_h\},\, A_\ell = a\Big) \;\propto\; \underbrace{\phi\Big(\tfrac{t-\theta_\ell}{\sigma_\ell}\Big)}_{\text{pre-selection density}} \times \underbrace{\mathbb{P}\Big(\textstyle\bigcap_h \{S_h = s_h\} \,\Big|\, T_\ell = t,\, A_\ell = a\Big)}_{\text{correction for selection}}$$

Splits are drawn independently at each node

$$\text{correction for selection} \;=\; \prod_{h=1}^{d} \Lambda_h(t)$$

$$\Lambda_h(t) \equiv \Lambda_h(t; a) \;=\; \mathbb{P}(S_h = s_h \mid Y = y(t,a)) \;=\; \frac{\exp\big(\tau_{p_h}^{-1} G_{p_h}(s_h;\, y(t,a))\big)}{\sum_{s' \in \mathcal{K}(p_h)} \exp\big(\tau_{p_h}^{-1} G_{p_h}(s';\, y(t,a))\big)}$$

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Inference on leaf nodes

An exact pivot, in closed form

$$\mathrm{Pivot}(T_\ell, A_\ell;\, \theta_\ell) \;=\; \frac{\displaystyle\int_{-\infty}^{T_\ell} \phi\big(\tfrac{t-\theta_\ell}{\sigma_\ell}\big) \prod_{h\in[d]} \Lambda_h(t; A_\ell)\, dt}{\displaystyle\int_{-\infty}^{+\infty} \phi\big(\tfrac{t-\theta_\ell}{\sigma_\ell}\big) \prod_{h\in[d]} \Lambda_h(t; A_\ell)\, dt}$$

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The randomization shapes the correction

	randomization	selection correction	data
Tree-Values (Neufeld et al., 2022)	none	hard truncation $\mathbb{1}(t \in M_t)$	Gaussian only
RRT (Bakshi et al., 2024)	Gaussian noise on gains	no closed form; numerical approximation	Gaussian only
iCART (this talk)	**softmax (exp. mechanism)**	**closed-form softmax product $\prod_h \Lambda_h$**	**distribution-free asymptotics (e.g. Bernoulli)**

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Toward distribution-free inference

The pivot needs only that T_ℓ is (asymptotically) Gaussian: a CLT, *not* a model for the data.

Gaussian data: **exact**. General data, e.g. Bernoulli (classification trees): *same* pivot, *same* $\prod_h \Lambda_h \Rightarrow$ asymptotically $\mathrm{Unif}(0,1)$, via a Lindeberg-type argument.

Why the softmax correction transfers

$\Lambda_h(t) > 0$ for *every* t, so no truncation boundaries; $\log \Lambda_h$ Lipschitz, bounded derivatives (Props. S4.2–S4.3), so the CLT carries through the correction.

Valid even as selection probabilities $\to 0$: no rare-event assumptions (vs. bounded away from 0 (Tian and Taylor, 2018); specific rates (Rasines and Young, 2023; Panigrahi, 2023))

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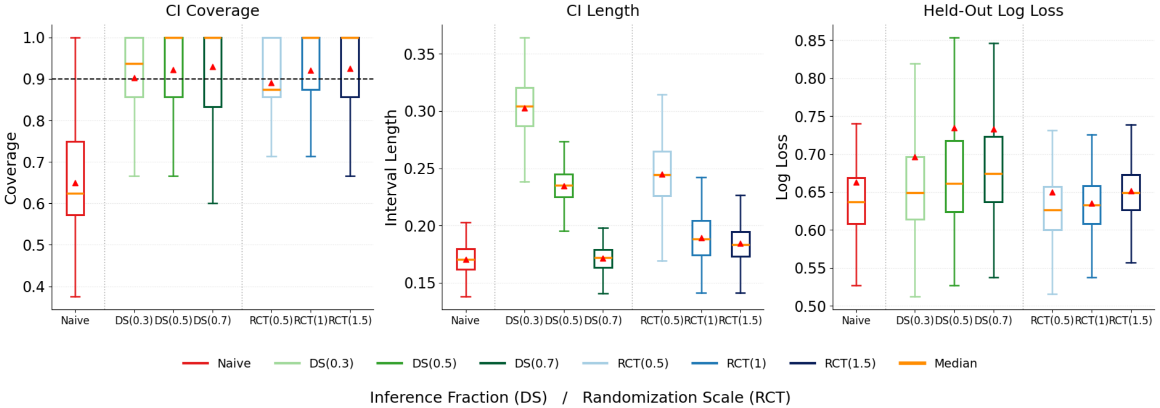
Simulation Synthetic data results



DS(f): data splitting, inference fraction f; RCT(τ) = **iCART** at randomization scale τ. iCART covers at 0.9 with short CIs *and* low log-loss — DS gets one or the other.

Takeaways

1. τ is a **user-facing dial**: at moderate τ, accuracy $\approx$ greedy CART
2. **beats data splitting**: all n used twice, validly $\Rightarrow$ shorter CIs, better fits
3. **exact** for Gaussian data, **distribution-free** asymptotics, a **first** for classification trees

powerful conditional inference + **near-optimal** test prediction performance

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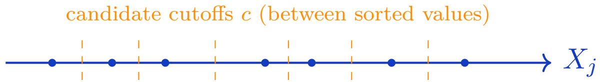
Candidate splits $\mathcal{K}(\mathcal{P})$

- **Continuous X_j:** threshold splits $\{X_j \le c\}$; cutoffs c = midpoints of the *sorted distinct* values in $\mathcal{P}$ $\Rightarrow$ up to $n_{\mathcal{P}} - 1$ per predictor.
- **Categorical X_j (q levels):** up to $2^{q-1} - 1$ binary partitions.



Size of the search

$|\mathcal{K}(\mathcal{P})| = O(n_{\mathcal{P}}\, p)$ for continuous features, growing with samples-in-region $n_{\mathcal{P}}$ and dimension p.

scikit-learn: `splitter="best"` scans every cutoff that *respects* `min_samples_leaf` (R's `minbucket`); `ExtraTrees`/`"random"` samples cutoffs, and histogram methods bin features ($\sim$`max_bins`) for large n, p. Our softmax reuses the argmax's gains, so randomization adds no cost.

The gain function: any *smooth* impurity works

$$G_{\mathcal{P}}(s';y) = I_{\mathcal{P}} - \left[\frac{n_{\mathcal{P}(s')}}{n_{\mathcal{P}}} I_{\mathcal{P}(s')} + \frac{n_{\mathcal{P}\setminus\mathcal{P}(s')}}{n_{\mathcal{P}}} I_{\mathcal{P}\setminus\mathcal{P}(s')}\right]$$

region impurity $I_{\mathcal{R}} = I(\hat{\pi}_{\mathcal{R}})$ (classification) / $I(\bar{y}_{\mathcal{R}})$ (regression).

Classification ($\hat{\pi}_{\mathcal{P}}$ = class-1 rate in $\mathcal{P}$)

Measure	$I(\hat{\pi}_{\mathcal{P}})$
Misclass.	$1 - \max(\hat{\pi}_{\mathcal{P}}, 1-\hat{\pi}_{\mathcal{P}})$ *
Gini	$2\hat{\pi}_{\mathcal{P}}(1-\hat{\pi}_{\mathcal{P}})$
Cross-entropy	$-\hat{\pi}_{\mathcal{P}} \log \hat{\pi}_{\mathcal{P}} - (1-\hat{\pi}_{\mathcal{P}}) \log(1-\hat{\pi}_{\mathcal{P}})$

Regression ($\bar{y}_{\mathcal{P}}$ = mean in $\mathcal{P}$)

Measure	$I(\bar{y}_{\mathcal{P}})$
Variance / MSE	$\frac{1}{n_{\mathcal{P}}} \sum_{i\in\mathcal{P}} (y_i - \bar{y}_{\mathcal{P}})^2$
MAE	$\frac{1}{n_{\mathcal{P}}} \sum_{i\in\mathcal{P}} \lvert y_i - \mathrm{med}_{\mathcal{P}} \rvert$

Agnostic to the choice of G (Assumption 4)

valid for any impurity $I : (0,1) \to \mathbb{R}$ that is bounded and *thrice-differentiable* with bounded derivatives: Gini, cross-entropy ✓; * misclassification's kink at $\frac{1}{2}$ needs a smooth surrogate.

One formula, three names

$$\mathbb{P}(S = s) \;\propto\; \exp\Big(\text{score}(s)\,/\,\text{temperature}\Big)$$

	score	temperature	community
Gibbs / Boltzmann	$-E(s)$ (energy)	T	statistical physics
Softmax	z_s (logit)	T	ML / LLM sampling
Exponential mechanism	$u(s)$ (utility)	$2\Delta/\varepsilon^{\mathrm{dp}}$	differential privacy
Our split rule	$G_{\mathcal{P}}(s;y)$ (gain)	$\tau_{\mathcal{P}}$	**this talk**

Gumbel–max: sampling = argmax after adding i.i.d. Gumbel noise to the scores. McSherry and Talwar (2007); Awan et al. (2019); Karwa and Slavković (2016).

We neither require nor seek privacy or stability, only **valid inference**.

A free byproduct: differential privacy

The split rule is the exponential mechanism of McSherry and Talwar (2007), so it is **differentially private** in the responses y (design held fixed).

$$\Delta_{\mathcal{P}} := \max_{s' \in \mathcal{K}(\mathcal{P})} \max_{y \sim y'} \left| G_{\mathcal{P}}(s'; y) - G_{\mathcal{P}}(s'; y') \right|$$

gain *sensitivity* at $\mathcal{P}$; $y \sim y'$: two response vectors differing in one coordinate.

- one split at $\mathcal{P}$ is $\varepsilon_{\mathcal{P}}^{\mathrm{dp}}$-DP with $\varepsilon_{\mathcal{P}}^{\mathrm{dp}} = 2\Delta_{\mathcal{P}}/\tau_{\mathcal{P}}$;
- the d splits on a root-to-leaf path compose to $\left(2\sum_{h=1}^{d} \Delta_{p_h}/\tau_h\right)$-DP.

smaller temperature $\tau_{\mathcal{P}}$ = greedier fit = *larger* (weaker) privacy budget , so temperatures stay bounded away from 0 (Assumption 5). Privacy is a byproduct, not a goal.

How good is the sampled selection?

A *general* discrete softmax: candidates $s' \in \mathcal{K}$, gain $G(s')$, temperature τ,

$$\mathbb{P}(S = s) = \frac{e^{G(s)/\tau}}{\sum_{s' \in \mathcal{K}} e^{G(s')/\tau}}, \qquad G^\star := \max_{s' \in \mathcal{K}} G(s').$$

Near-optimal selection (exponential-mechanism utility)

$$\mathbb{P}\Big(G^\star - G(S) \;\geq\; \tau\,(\log|\mathcal{K}| + t) \Big) \;\leq\; e^{-t} \qquad \forall\, t > 0.$$

- expected regret $\lesssim \tau \log|\mathcal{K}|$: the sampled winner is within $O(\tau \log|\mathcal{K}|)$ of the greedy optimum;
- hitting the optimum: $\mathbb{P}(S = s^\star) \geq \left(1 + (|\mathcal{K}| - 1)\, e^{-\delta/\tau}\right)^{-1}$, runner-up gap $\delta = G^\star - \max_{s' \neq s^\star} G(s')$;
- $\tau \to 0$ recovers the greedy argmax (Proposition 3.1).

applies verbatim to **tree splits** ($\mathcal{K} = \mathcal{K}(\mathcal{P})$), **model / winner selection**, or *any* discrete softmax. McSherry and Talwar (2007); cf. arXiv:2607.18545 §2.2.

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Why softmax, not Gaussian noise?

Gumbel–max: the softmax *is* additive randomization

drawing from the softmax = adding i.i.d. Gumbel noise to the gains, then argmax (Prop. 3.2): the additive-noise paradigm, adapted to a *discrete* selection.

Under *any* additive noise ω, the weight entering the pivot is an orthant probability; write $g_{s'} := G_{p_h}(s'; y(t,a))$:

$$\Lambda_h(t;a) = \mathbb{P}\big(g_{s_h} + \omega_{s_h} \;>\; g_{s'} + \omega_{s'} \quad \forall\, s' \in \mathcal{K}(p_h)\big)$$

Gumbel noise

collapses to the softmax: closed form, $O(|\mathcal{K}(p_h)|)$ per node; log-weight is **Lipschitz** with bounded derivatives up to third order (Props. S4.2–S4.3), exactly the regularity the asymptotic theory requires.

Gaussian noise

no closed form: isotropic case $\int \phi(u) \prod_{s' \neq s_h} \Phi\big(u + (g_{s_h} - g_{s'})/\sigma\big) du$, i.e. quadrature at *every node and grid point*; correlated: a $(|\mathcal{K}(p_h)| - 1)$-dim orthant probability. Regularity only via the integral.

References I

Ackley, D. H., Hinton, G. E., and Sejnowski, T. J. (1985). A learning algorithm for boltzmann machines. *Cognitive science*, 9(1):147–169.

Awan, J., Kenney, A., Reimherr, M., and Slavković, A. (2019). Benefits and pitfalls of the exponential mechanism with applications to hilbert spaces and functional pca. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 374–384. PMLR. ICML 2019.

Bakshi, S., Huang, Y., Panigrahi, S., and Dempsey, W. (2024). Inference with randomized regression trees. *arXiv preprint arXiv:2412.20535*.

Breiman, L. (2001). Statistical modeling: The two cultures (with comments and a rejoinder by the author). *Statistical science*, 16(3):199–231.

Breiman, L., Friedman, J., Stone, C., and Olshen, R. (1984). *Classification and Regression Trees*. Taylor & Francis.

Dong, J., Roth, A., and Zhou, H. (2020). Optimal differential privacy composition for exponential mechanisms. In *Proceedings of the 37th International Conference on Machine Learning*, pages 2597–2607.

Dwork, C. and Roth, A. (2014). *The Algorithmic Foundations of Differential Privacy*, volume 9 of *Foundations and Trends in Theoretical Computer Science*. Now Publishers Inc.

Holtzman, A., Buys, J., Du, L., Forbes, M., and Choi, Y. (2020). The curious case of neural text degeneration. In *International Conference on Learning Representations (ICLR)*.

References II

Karwa, V. and Slavković, A. (2016). Inference using noisy degrees: Differentially private β-model and synthetic graphs. *Annals of Statistics*, 44(1):87–112.

Lindeberg, J. W. (1922). Eine neue herleitung des exponentialgesetzes in der wahrscheinlichkeitsrechnung. *Mathematische Zeitschrift*, 15:211–225.

Maddison, C. J., Tarlow, D., and Minka, T. (2014). A* sampling: A fast sampling method for discrete distributions. In *Advances in Neural Information Processing Systems*, volume 27, pages 3086–3094. Curran Associates, Inc.

McSherry, F. and Talwar, K. (2007). Mechanism design via differential privacy. In *48th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 94–103.

Neufeld, A. C., Gao, L. L., and Witten, D. M. (2022). Tree-values: selective inference for regression trees. *Journal of Machine Learning Research*, 23(305):1–43.

Panigrahi, S. (2023). Carving model-free inference. *The Annals of Statistics*, 51(6):2318–2341.

Quinlan, J. R. (2014). *C4. 5: programs for machine learning*. Elsevier.

Rasines, D. G. and Young, G. A. (2023). Splitting strategies for post-selection inference. *Biometrika*, 110(3):597–614.

Tian, X. and Taylor, J. (2018). Selective inference with a randomized response. *The Annals of Statistics*, 46(2):679–710.

Zhang, T., Lee, H., and Lei, J. (2024). Winners with confidence: Discrete argmin inference with an application to model selection. *arXiv preprint arXiv:2408.02060*.